

COMPARATIVE ECONOMIC ANALYSIS OF SELECTED TIMBER SPECIES IN OGUN STATE, NIGERIA.

CHAPTER ONE

INTRODUCTION

1.1 Background of The Study

Timber is one of the most valuable forest products. In Nigeria, it is used as raw material for construction, furniture, doors and window frames, packaging, and many household and industrial purposes. Beyond these uses, the timber trade also supports livelihoods by providing income and jobs for loggers, sawmill operators, transporters, plank sellers, and other traders along the value chain. Ogun State, in south-western Nigeria, takes part in this chain through its forest resources and the many towns with sawmills and plank markets that serve both local buyers and customers from neighbouring states.

Timber trading usually involves buying logs or planks, transporting them to market, and often processing, storing, and selling them. Each stage adds cost: purchase price, transport, loading and offloading, sawing or processing, labour, storage, and levies. A trader's earnings depend on the difference between total costs and sales revenue. Timber species also vary in demand, durability, size, and availability, so a higher selling price does not always mean higher profit, since some species cost more to handle.

Since 2020, several studies across Nigeria have examined the economics of timber trading using different methods. In Osun State, Babatunde et al. (2020) assessed profitability, marketing efficiency, and value addition in Ife East LGA, while Falana et al. (2023) analyzed sawnwood marketing in Isokan LGA and found the business worthwhile across eight major species. In Ogun State, Babatunde et al. (2022) used gross margin, net income, and cost-benefit analysis to study sawnwood production in Ijebu-Ode, and Babatunde et al. (2020)

evaluated the profitability of non-timber forest products at Arakanga Forest Reserve, Abeokuta. In Oyo State, Babatunde et al. (2021) reported on timber marketing in selected sawmills in Ibadan North LGA, and Babatunde et al. (2026) found wood marketing in Ikire to be profitable. Other studies have compared species directly. Sambe et al. (2022) calculated gross margins for several species in Lokoja, Kogi State. Nwandu et al. (2021) analysed sawn timber marketing in Sapele, Delta State, using gross margin and regression. Adedire et al. (2023) in Plateau State used correlation and multiple regression to link timber sales with costs, demand, stored stock, and spoilage.

Simple profitability indicators can summarise how well a trading business is doing. Gross margin is total revenue minus total variable costs. Net profit is total revenue minus total costs, both fixed and variable. The benefit-cost ratio (BCR) is total benefit divided by total cost, and it shows the return per naira spent. Calculating these measures by species, and combining them with tools such as correlation and regression, helps identify the best-performing species and the factors that affect timber prices and trader profits.

1.2 Problem Statement

Ogun State's timber traders deal in a wide variety of wood species, but there is little current data on which species are most frequently sold in the markets, how much it costs to trade each species, what each species sells for, and which species yields the highest financial return. When choosing which species to purchase and how much to charge, many traders rely more on habit and experience than on documented costs and sales data. Because of this, a species with a high selling price could be preferred even if its high expenses result in a smaller profit than those of other species.

For Ogun State's markets, it is also difficult to quantify the elements that lead to variations in pricing and profits between species and between dealers, such as transportation costs,

quantity traded, quality, season, market location, and trader experience. Although their impact on prices and profit in Ogun State markets has not been measured, constraints such as high transportation costs, scarcity of preferred species, price fluctuation, and levies have been reported elsewhere (Falana et al., 2023; Sambe et al., 2022; Nwandu et al., 2021).

Instead of comparing multiple wood species across specific timber markets, current research in the State has focused on non-timber forest products (Babatunde et al., 2020b) or a single sawmill location (Babatunde et al., 2022). Therefore, a current, species-by-species comparison of costs, selling prices, and profitability in certain Ogun State wood markets is required, along with an examination of the variables influencing prices and profitability utilizing gross margin, net profit, and benefit-cost ratio.

1.3 Aim and Objectives

The aim of a comparative economic analysis of selected timber species traded in selected timber markets in Ogun State, Nigeria is to provide evidence on their costs, selling prices and profitability, and on the factors that influence them.

The specific objectives are to:

- i. identify the major timber species traded in selected timber markets in Ogun State;
- ii. determine and compare the costs and selling prices of the selected timber species;
- iii. compare the profitability of the selected timber species using gross margin, net profit and benefit-cost ratio; and
- iv. identify the factors influencing the prices and profitability of the selected timber species.

Together, these objectives show which timber species are traded most, how much they cost and sell for, which of them give the best returns, and what drives those differences.

1.4 Significance of The Study

This study will give timber traders in Ogun State accurate facts on the costs and returns of the numerous species they trade. As a result of traders, traders, traders, traders, traders, traders, traders, traders, experience, stock, stock, stock, decisions, decisions, decisions, decisions, information, decisions, decisions, decisions, decisions, prices, and so as well as well as well as well as well as well as well as well as well. It will also allow new entrants and investors who seek to enter the timber sector to know which species offer greater returns.

The findings will support forestry agencies and policymakers by providing evidence on the most frequently traded timber species and the factors influencing their prices. This information can inform regulations, taxation, market levies, and the promotion of tree species that are in high demand, a concern also identified by Adedire et al. (2023) in Plateau State.

This study will further contribute to research on timber marketing and forest economics in Nigeria. It will also provide a useful reference for students and researchers conducting related studies in other states or examining other forest products.

1.5 Scope of The Study

This study is limited to the economic analysis of selected timber species traded in selected timber markets in Ogun State, Nigeria. Timber dealers and plank vendors working in the designated markets make up the study population. A standardized questionnaire covering these merchants' personal traits, the species they trade, their expenses, their sales, and the variables influencing their prices and profits will be used to gather primary data from them.

The species traded in the chosen marketplaces, their expenses and selling prices over a specified time period, and their profitability as determined by gross margin, net profit, and benefit-cost ratio will all be included in the analysis. In accordance with the techniques employed by Adedire et al. (2023) and Nwandu et al. (2021), descriptive statistics, correlation analysis, and regression analysis will be utilized to characterize the data and

determine the factors influencing pricing and profitability. The study's conclusions are primarily applicable to the markets and time period under investigation; it does not address the biological state of forests, timber harvesting within the forest, or the timber trade beyond the chosen markets.

1.6 Motivation of The Study

In Nigeria, timber is still a vital building and furnishing resource, and its prices are influenced by supply, demand, quality, and transportation costs. There is little recent data on how the various species compare in terms of cost, price, and profit in Ogun State's markets, despite the fact that traders must adapt to these changes. Cost and sales data can be transformed into valuable conclusions about profitability and limits, according to recent research conducted in various regions of the southwest and abroad (Babatunde et al., 2020a; Babatunde et al., 2021; Babatunde et al., 2026; Falana et al., 2023; Sambe et al., 2022). This encouraged the present work to extend such analysis to a species-by-species comparison across selected timber markets in Ogun State.

This study is also motivated by the need to move from general statements about the timber business to measured evidence. By combining profitability analysis with statistical methods, this study aims to provide results that traders, policy makers and researchers can use with confidence to understand the economics of timber species in Ogun State.

CHAPTER TWO

LITERATURE REVIEW

2.1 Theoretical Background of Timber Marketing and Profitability

Timber marketing covers the movement of timber from the forest or sawmill to the final user. It includes buying logs or planks, transporting them, processing or sawing, storing and selling. The people involved include loggers, sawmill operators, wholesalers and plank sellers, and many of them operate as small private businesses (Akinola, 2025; Babatunde et al., 2022). Because each of these steps involves a cost, the economic performance of a timber trader depends on the relationship between what is spent and what is earned from sales.

Studies on timber and wood marketing in Nigeria rely mainly on cost and return analysis, also called budgetary analysis. In this approach, costs are separated into variable costs, which change with the volume of business, and fixed costs, which stay the same over the period. Variable costs include the purchase of timber, transport, labour, fuel, processing and taxes, while fixed costs include items such as depreciation on saw machines, generating sets and vehicles (Babatunde et al., 2026). Revenue is then compared with these costs using indicators such as gross margin, net income and returns on investment (Nwandu et al., 2021; Babatunde et al., 2026).

Wood trade has also been studied as a value chain. Ali et al. (2023) examined the fuelwood value chain in northern Nigeria from economic, environmental and social points of view, and used the distribution of profit along the chain as one of their indicators. Although fuelwood is not the same product as sawn timber, this view is useful because it shows that profitability should be studied together with how benefits are shared among traders and whether the trade can be sustained.

When the aim is to explain why prices or profits differ between traders, researchers commonly estimate regression models in which sales, profit or margin is the dependent

variable, and costs and trader characteristics are the explanatory variables (Adedire et al., 2023; Agbugba et al., 2021; Nwandu et al., 2021). Some studies go further and use efficiency models. Babatunde et al. (2026), for example, applied stochastic frontier analysis to measure how well wood marketers used their inputs. These approaches form the theoretical basis for the methods proposed in this study.

2.2 Timber Species and Timber Markets in Nigeria

2.2.1 Timber Supply and Species Availability

The demand for wood and wood products in Nigeria keeps rising, and this puts pressure on the forest estate. Akomolede et al. (2022) studied saw millers in Odigbo Local Government Area of Ondo State and identified about twenty indigenous timber species in use. Many of them were classified as vulnerable or endangered, and the authors concluded that the rate of exploitation had led to a decline in the number of indigenous species. Olajiire-Ajayi et al. (2021) recorded timber logs leaving Shasha Forest Reserve in Osun State over twenty-eight weeks, assessing 13,944 logs of twenty-one species, and found that *Ricinodendron heudelotii* was the most frequently exploited. These findings suggest that supply, and therefore the availability and price of particular species, can change as some species become scarce.

The species traded also differ from one market to another. Falana et al. (2023) reported eight major species in the sawnwood markets of Isokan Local Government Area of Osun State. In Akure, Akinola (2025) found that afara (*Terminalia superba*), mahogany and obeche were among the most commonly purchased species. Sambe et al. (2022) compared seven species in Lokoja, and Adedire et al. (2023) found that mahogany (*Khaya senegalensis*) was the species mainly demanded by consumers in the markets of Plateau State. Since the species mix is not the same everywhere, a species-by-species study of Ogun State must first identify which species are actually traded in the selected markets, which is the first objective of this study.

Fuelwood is another wood product on which pressure is placed. Audu and Meleми (2023) noted that rural and urban poor households in Nigeria depend mainly on fuelwood for energy, and that this raises consumption and affects the environment. This competing demand for wood adds to the pressure on tree resources from which timber is also drawn.

2.2.2 Timber Markets and Traders

Timber is mostly sold as sawn planks in markets close to sawmills or along major roads, and traders are usually small private operators. Babatunde et al. (2022) collected data from plank sellers in fifteen sawmills out of sixty in Ijebu-Ode Local Government Area of Ogun State, and Akinola (2025) surveyed ten sawmills and eighty plank sellers in Akure. Falana et al. (2023) found that most of the respondents in Isokan were men, and that many entered plank marketing through training and started with bank loans. Babatunde et al. (2020a) surveyed timber marketers and furniture firms in Ife East Local Government Area of Osun State and examined their marketing efficiency, profitability and the level of value addition.

Market structure also affects how traders earn. Sambe et al. (2022) used a Gini coefficient for the Lokoja markets and concluded that the timber market was imperfectly competitive, with a structure tending towards monopoly. Such findings show that market organisation can influence prices, and that studies of price and profit should consider the market in which a trader operates.

2.3 Measures of Profitability

The most common measures of profitability in the timber studies reviewed are gross margin, net income or net profit, and ratios of return to cost. The gross margin is calculated as total revenue minus total variable cost (Nwandu et al., 2021; Babatunde et al., 2026). The net profit is total revenue minus total cost, where total cost includes both variable and fixed costs.

The benefit-cost ratio divides total benefit by total cost, and a value above one shows that every naira spent returns more than a naira.

Different studies report these measures in different ways. Babatunde et al. (2022) reported a gross margin of about ₦7.65 million, a net income of about ₦6.96 million and a cost-benefit ratio of 3.32 for sawn wood production in Ijebu-Ode. Agbugba et al. (2021), studying wood charcoal production in Rivers State, reported a cost-benefit ratio of 1:3, together with a gross margin and net income, and concluded that the business was profitable. Babatunde et al. (2026) calculated return on investment values between 2.4 and 3.1 for wood marketing in Ikire, while Nwandu et al. (2021) reported a return on investment of 1.037 for sawn timber marketers in Sapele. Because the definitions and units are not always the same, each measure used in the present study will be defined clearly so that the results can be compared fairly.

Only a few studies calculate these measures for individual species. Sambe et al. (2022) worked out income per truck for seven timber species in Lokoja and found that *Daniellia oliveri* recorded the highest mean monthly return, while *Ficus exasperata* had the lowest mean monthly gross margin. This kind of species-level comparison is what the present study applies to selected timber species in Ogun State.

2.4 Factors Influencing Timber Prices and Profitability

2.4.1 Cost Factors

Transport cost is one of the most frequently reported factors. Nwandu et al. (2021) found that selling price and the age of the sawmill were positively related to marketing margin, while transport cost was negatively related and statistically significant. Falana et al. (2023) and Babatunde et al. (2026) also listed transportation and bad roads among the main constraints faced by traders. In the supply model of Babatunde et al. (2022), the distance of the sawmill from log sources, the cost of processing materials and the cost of labour were significant

determinants of sawn wood supply, and Babatunde et al. (2026) found that labour, capital and processing cost significantly increased output.

Power supply is another cost-related factor. Akinola (2025) reported that unreliable power, together with high labour and equipment costs, affected the profitability and sales of sawmills and plank sellers in Akure. In Ikire, inadequate electricity and high transport cost were the most widely reported problems, each named by almost all respondents (Babatunde et al., 2026).

2.4.2 Market and Trader Characteristics

The characteristics of the trader and the business have been linked to performance. In Ijebu-Ode, the availability of credit facilities and electricity was a significant determinant of supply (Babatunde et al., 2022), and in Ikire inadequate credit was reported by most marketers (Babatunde et al., 2026). Akinola (2025) found a significant association between years of experience and the barriers faced by plank sellers, but no significant association between education and those barriers. Falana et al. (2023) reported a significant relationship between the socio-economic characteristics of respondents and the constraints they faced. Such results justify including experience, quantity traded and credit access among the possible explanatory variables in this study.

2.4.3 Demand, Season and Policy

Demand and seasonal factors also matter. In Plateau State, Adedire et al. (2023) used Pearson correlation and multiple linear regression and found that timber sales were positively correlated with costs, demand, stored timber and spoiled timber, and that these variables significantly affected sales. Nwandu et al. (2021) reported that the type of species sold, seasonality and transport cost were the main factors that influenced the price of sawn timber. Falana et al. (2023) listed price fluctuation in purchase, government tariffs and scarcity of timber as constraints, and Sambe et al. (2022) reported low patronage, scarcity of demanded

species and high tax as the major challenges in Lokoja. Government policy was also reported as a constraint in Akure (Akinola, 2025) and in Ife East (Babatunde et al., 2020a).

2.5 Related Work

Babatunde et al. (2020a) studied the profitability, marketing efficiency and value addition of the timber industry in Ife East Local Government Area, Osun State. They selected fifteen sawmills at random and distributed one hundred and fifty questionnaires to timber marketers and furniture firms. The study found that benches added the highest value to the industry, and that government policy, inadequate market facilities, lack of credit, poor power supply and high transport cost were the main constraints. In the same year, Babatunde et al. (2020b) examined the profitability of non-timber forest products collected in Arakanga Forest Reserve, Abeokuta, Ogun State, showing that profitability studies of forest products have been carried out in Ogun State, although for non-timber products.

Babatunde et al. (2021) reported on the marketing of timber in selected sawmills in Ibadan North Local Government Area of Oyo State and found the business to be profitable. Nwandu et al. (2021) analysed sawn timber marketing in Sapele, Delta State, using descriptive statistics, gross margin analysis and a regression model of profit. They found the business profitable and reported that selling price, transport cost and the age of the sawmill were significant in the regression, while the type of species sold, seasonality and transport cost were the main constraints. Agbugba et al. (2021) used descriptive statistics, gross margin and multiple regression to study wood charcoal production and marketing in Rivers State, using thirty producers and one hundred marketers, and found both activities profitable.

Babatunde et al. (2022) studied sawn wood production in Ijebu-Ode Local Government Area, Ogun State. Out of 150 plank sellers, 143 questionnaires were retrieved. The study calculated gross margin, net income and a cost-benefit ratio of 3.32, and used regression to show that distance to log sources, credit facilities, cost of processing materials, cost of labour and

availability of power were significant determinants of supply. Sambe et al. (2022) analysed timber marketing in Lokoja, Kogi State, and calculated gross margins for several species. They found differences in returns between species and concluded that low patronage, scarcity of demanded species and high tax were the main challenges.

Akomolede et al. (2022) and Olajiire-Ajayi et al. (2021) focused on the supply side. Akomolede et al. identified the indigenous timber species used by saw millers in Odigbo, Ondo State, and showed that many were vulnerable or endangered, while Olajiire-Ajayi et al. measured timber exploitation in Shasha Forest Reserve, Osun State. Both studies recommended stronger monitoring and policy enforcement. Ali et al. (2023) and Audu and Melemi (2023) looked at wood used as fuel in northern Nigeria, the first from a value chain perspective and the second from the consumption side, and both stressed the pressure that wood use places on forests and the livelihood role of the trade.

Falana et al. (2023) analysed sawnwood marketing in Isokan Local Government Area, Osun State, and found eight major species on sale and a business that returned more than it cost. Adedire et al. (2023) studied sales, costs and demand in selected timber markets of Plateau State and showed, using correlation and multiple linear regression, that costs, demand, stored timber and spoiled timber were related to sales. Akinola (2025) examined the barriers to wood plank trading in Akure, Ondo State, using questionnaires from ten sawmills and eighty plank sellers and a chi-square test. The study identified government policy, unreliable power, high labour and equipment costs and limited wood availability as barriers. Most recently, Babatunde et al. (2026) used budgetary analysis and stochastic frontier analysis for ninety wood marketers in Ikire, Osun State, and found profitable businesses that operated at an average technical efficiency of 0.82.

2.6 Research Gap

The studies reviewed show that timber and wood marketing in south-western Nigeria and other parts of the country is generally profitable, and that costs, transport, power supply, credit, demand and policy affect performance. Most of them, however, treat timber traders as one group. Within Ogun State, the recent studies reviewed focus on a single sawmill area (Babatunde et al., 2022) or on non-timber forest products (Babatunde et al., 2020b). Studies that compare species directly, such as Sambe et al. (2022) and Adedire et al. (2023), were carried out in Kogi and Plateau States, and they do not use gross margin, net profit and benefit-cost ratio together with a regression analysis of price and profit in selected markets in Ogun State.

The present study therefore addresses this gap by identifying the major timber species traded in selected timber markets in Ogun State, comparing their costs, selling prices and profitability using gross margin, net profit and benefit-cost ratio, and identifying the factors that influence their prices and profitability.

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